Innovation, Imagination, And Emotional Computing

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A future trend I see is what I call "emotional computing." It's not technology for technology's sake; it's about technology making it easier to communicate, enjoy life, and care for our loved ones. More and more of our MIPS and FLOPS and megabytes are going to be poured into touching an emotional chord within us individually. A good example is Apple's iconic iPod. The genius of Steve Jobs is taking it from technology to an emotional experience.

An area where this is clearly expressed is in the healthcare industry. To share an example, a pioneering platform that was built specifically for healthcare is the MCA—Mobile Clinical Assistant. The lightweight, spill-resistant, drop-tolerant, and easily disinfected MCA allows nurses to access up-to-the-minute patient records and to document a patient's condition instantly, enhancing clinical workflow while reducing the staff's administrative workload. Some of the features designed to ease the nurse's daily workload include: wireless connectivity to access up-to-date secure patient information and physician's orders; radio frequency identification (RFID) technology for easy, rapid user logon; a digital camera to enhance patient charting and progress notes, to keep track of wounds as they heal; and Bluetooth technology to help capture patient vital signs. By better connecting clinicians to comprehensive patient information on a real-time basis, this technology innovation helps usher in improvements in the fundamental areas of healthcare quality, access and cost. Nurses who use a mobile clinical assistant don't want to give it up. We need to develop more products that create that kind of emotional tie with people.

Another technology development that will change consumers' lives in the foreseeable future is that of "context aware technology." It's bringing *Minority Report* science fiction to life. In the movie, Tom Cruise walked into a Gap store and the advertisements at the entrance changed in response to his presence. That's this notion of context awareness and being able to leverage resources accordingly. The displays adapt to his needs and his purchasing history. The futuristic vision is when we will have portable devices with context awareness. These devices would constantly adapt to their environment, knowing if they are at home, at work, in the car, and whether it's day or night. That kind of vision is not just imagination or fiction, it is becoming a reality. In our research labs, we are working on sensor requirements for this kind of ubiquitous computing so that devices can react appropriately to their environment and what people are doing, and be more useful to the people who own them. While these innovative technologies will provide opportunities to grow into new areas, they will have disruptive and unique demands from the hardware.

Demand For Integration

Chip integration is the future. Today's hardware platforms will continue to be integrated into fewer and fewer chips in the next few years. The platform is moving into the processor, and what was defined as a platform just a year ago is going to be integrated onto a single chip in the next few years. Integration brings a number of benefits, including higher performance, reduced power consumption, lower cost, and reduced size. Things are changing as we move into the System on a Chip (SoC) world. Future applications for SoC devices will need increased levels of customisation and faster design cycles. Today, CPUs take about four years from inception to production, but to serve the consumer market, we must now consider significantly shortening that time.

Petition For Performance

The "Era of Tera"—with Teraflops of computing performance crunching through Terabytes of people's data–is in the visible future. When we saw the world's first programmable processor that delivered Teraflops performance with remarkable energy efficiency, we were peeping into the not-so-distant future. A single, 80-core chip not much larger than the size of a fingernail, delivering supercomputer-like performance, while using less electricity than most of today's home appliances. This "Tera-scale computing" research innovation is aimed at delivering Teraflops—or trillions of calculations per second—performance for future PCs and servers. Tera-scale performance, and the ability to move terabytes of data, will play a pivotal role in future computers with ubiquitous access to the Internet by powering new applications for education and collaboration, as well as enabling the rise of high-definition entertainment on PCs, servers, and handheld devices. For example, artificial intelligence, instant video communications, photo-realistic games, multimedia data mining and real-time speech recognition—once deemed as science fiction in *Star Trek* shows—could become everyday realities.

It's a brave new world ahead. We are accelerating the pace of innovation to transform imagination into reality and to evolve technology into an emotional experience. ■

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